Deep Learning Mini Project Synopsis

Illegal Surface Mining Detection Using Satellite Imagery

Group Details

Panel C

Batch C1

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Abstract

Illegal surface mining has become a serious environmental problem leading to issues like deforestation, soil damage, water pollution and description of ecosystems. Traditional monitoring methods rely on manual infection, which is not only time consuming but also ineffective when dealing with large or remote areas.

With advancements in satellite technology and artificial intelligence, it is now possible to automate such monitoring processes. This project presents an AI driven system that detects possible illegal surface mining activities using satellite imagery. The system takes satellite images as input and processes them through image processing techniques followed by feature extraction and classification using deep learning.

Due to the lack of directly labelled illegal mining datasets, the project uses the EuroSAT dataset as an alternative. Certain land use categories such as industrial areas and agricultural land are interpreted as potential mining indicators. a MobileNetV2-based convolutional neural network (CNN) is used for efficient image classification.

The system also includes a simple user interface where users can upload images and receive predictions such as “mining detected” or “non mining” along with confidence scores. The results are displayed and stored for future reference and analysis. Overall, this project highlights how AI and satellite data can be combined to support environmental monitoring and detection systems.

Keywords: Satellite Imagery, CNN, MobileNetV2, Remote Sensing, Environmental Monitoring

1. Introduction

Illegal surface mining is a growing concern across many parts of the world, as it causes significant environmental and economic damage. It leads to deforestation, soil erosion, water contamination and destruction of natural habitats. Detecting such activities is challenging, especially when they occur in remote or inaccessible areas where manual monitoring is difficult.

Satellite imagery offers a practical solution as it allows continuous observation of large geographical regions. However, analysing these images manually is not efficient and can lead to errors when dealing with vast datasets.

This is where artificial intelligence, particularly deep learning, becomes useful. Deep learning models can automatically identify patterns and images and classify different types of land use.

These technologies have already been applied in areas such as deforestation detection, urban expansion monitoring, and environmental analysis.

In this project, a simplified yet effective system is developed to detect possible mining activities using satellite images. Instead of relying on the specialised mining dataset, which is difficult to obtain, a publicly available data set is used the system classifies land use patterns and then applies a mapping logic to identify areas that may indicate mining activity.

This project is important because it combines:

- remote sensing,
- deep learning,
- environmental sustainability,
- and intelligent monitoring systems.

It also provides a strong academic foundation for future expansion into more advanced real-world mining surveillance applications.

2. Literature Review Table

Recent research shows a growing interest in using **remote sensing and deep learning** for environmental monitoring, land-cover classification, and mining detection.

Author(s)	**Title**	**Methods Used**	**Key Finding**	**Advantage**	**Disadvantage**
Giang et al. (2020)	U-Net Convolutional Networks for Mining Land Cover Classification Based on High-Resolution UAV Imagery	U-Net, semantic segmentation	Classified mining land cover effectively	High accuracy for land classification	Needs high-resolution labeled data

Sujatha et al. (2025)	Geo-AI for Environmental Monitoring: Detection of Landstrips to Support Illegal Mining Surveillance	YOLOv8-OBB, SAR + Optical fusion	Detected illegal mining-related landstrips	Works well with multimodal imagery	Focuses on landstrips, not direct mining detection
Lemes Neto et al. (2026)	Towards SAR-Based Monitoring of Illegal Mining in the Brazilian Amazon Using CNNs	CNN, SAR image classification	Detected illegal mining in cloud-prone regions	Useful in cloudy areas using SAR	Less effective for small disturbances
Murdaca et al. (2023)	A Semi-Supervised Deep Learning Framework for Change Detection in Open-Pit Mines Using SAR Imagery	Semi-supervised learning, SAR change detection	Detected mine changes effectively	Good for monitoring land changes	Requires multi-temporal images
Mehta et al. (2024)	Semantic Segmentation of Optical Satellite Images for the Illegal Construction Detection Using Transfer Learning	U-Net, transfer learning	Detected illegal structures using satellite images	Good use of transfer learning	Focused on construction, not mining

Summary of Literature:

From the reviewed papers, the following observations can be made:

- Deep learning is highly effective for analyzing remote sensing imagery.

- **U-Net, CNNs, and YOLO-based models** are commonly used for land classification and object detection.
- Both **optical** and **SAR imagery** are useful, each with different strengths.
- Most advanced works focus on **segmentation**, **object detection**, or **change detection**.
- Real-time environmental monitoring is becoming a major application area of Geo-AI.

3. Research Gap

- Although existing studies show promising results, there are still some practical gaps:
- Many systems rely on **specialized datasets** that are difficult for students and small research teams to access.
- Advanced methods such as **semantic segmentation on SAR or UAV imagery** are often computationally expensive.
- Few academic prototypes focus on creating a **simple end-to-end usable application** with upload, prediction, dashboard, and history tracking.
- There is limited work on building a **lightweight educational prototype** using transfer learning and public datasets for illegal mining approximation.

How This Project Fills the Gap

This project addresses the above gap by:

- using a **publicly available satellite image dataset (EuroSAT)**,
- applying a **lightweight pretrained CNN (MobileNetV2)**,
- converting land-use classification into **mining / non-mining detection** through rule-based mapping,
- and integrating the model into a **simple deployable web interface**.

So while this project is not a perfect real-world mining surveillance platform, it is a **practical, scalable, and academically valid prototype**

4. Objectives

The main goal of this project are:

- To design a system that can identify possible mining related regions using satellite images
- To apply preprocessing techniques to improve image quality for model training
- To use a pretrained MobileNetV2 model for classifying land use patterns
- To convert classification results into Mining or non mining categories using mapping logic

- To develop and easy to use interface for uploading images and weaving results
- To store prediction data and provide basic analytics through a dashboard
- To demonstrate how a i can contribute to environmental monitoring

5. Methodology / Proposed System

The proposed system follows a deep learning-based image classification pipeline to detect possible illegal surface mining regions from satellite imagery.

5.1 Overall Workflow

The overall flow of the system is:

Image Input → Preprocessing → CNN Feature Extraction → Classification → Mining Mapping → Output

This is the core working pipeline of the project.

5.2 Input Data

The model uses satellite images from EuroSAT dataset which includes classes such as:

- AnnualCrop
- Forest
- HerbaceousVegetation
- Highway
- Industrial
- Pasture
- PermanentCrop
- Residential
- River
- SeaLake

Since a dedicated illegal mining dataset is not easily available, this dataset is used as a **proxy land-use dataset**.

Mining Approximation Logic

Certain land-use categories are interpreted as mining-like or suspicious.

For example:

- **Industrial** → Mining-related
- **AnnualCrop** → Potentially disturbed land / mining-like in approximation
- Others → Non-Mining

This rule-based mapping enables the model to act as a simplified illegal mining detector.

5.3 Image Preprocessing

Before feeding images into the model, preprocessing is applied:

- **Resizing:** Images are resized to **128 × 128**
- **Normalization:** Pixel values are scaled between **0 and 1**
- **Data Augmentation:** Rotation, flipping, zooming, and shifting are applied

Why preprocessing is important

- It ensures all images are of the same size
- It improves training stability
- It reduces overfitting
- It helps the model generalize better

This stage is implemented using image preprocessing tools such as:

- ImageDataGenerator
- NumPy
- PIL

5.4 Deep Learning Model

The core of the proposed system is a **Convolutional Neural Network (CNN)** based on **MobileNetV2 Transfer Learning**.

Why MobileNetV2?

The system uses MobileNetV2, a lightweight CNN model that is fast and efficient. It is well suited for projects with limited computational resources and works well with transfer learning. The model extracts features such as textures patterns and spatial details from images and uses them to classify land types.

Model Architecture

The model includes:

- **MobileNetV2 pretrained base**
- Global pooling layer
- Dense layer
- Dropout layer
- Output classification layer

How it works

- The pretrained model extracts important image features such as:
- texture,
- color distribution,
- spatial patterns,
- structural patterns of land-use.

These features are then used to classify the input image into one of the land classes.

5.5 Training Phase

During training:

- The model learns from labeled satellite images
- It identifies patterns of:
  - industrial zones,
  - vegetation,
  - urban areas,
  - water bodies,
  - and other land-use classes

Training is performed using:

- TensorFlow / Keras
- model.fit()

The dataset is typically divided into:

- Training set
- Validation set
- Testing set

5.6 Prediction and Mining Detection Logic

After classification, the model returns class probabilities.

Example:

- Industrial → 0.85
- Forest → 0.07
- Residential → 0.03

The system then applies a **mapping rule**:

```
if predicted_class in ['Industrial', 'AnnualCrop']:
  result = "Mining Detected"
else:
  result = "Non-Mining"
```

This is the most important logic of the system because it converts **land-use classification** into **mining detection**.

5.7 User Interface

A simple frontend is built so the system is not just a model, but a usable application.

Features:

- Upload satellite image
- Run prediction
- Show predicted class
- Show mining / non-mining result
- Display confidence score
- Save result in database
- Show dashboard statistics

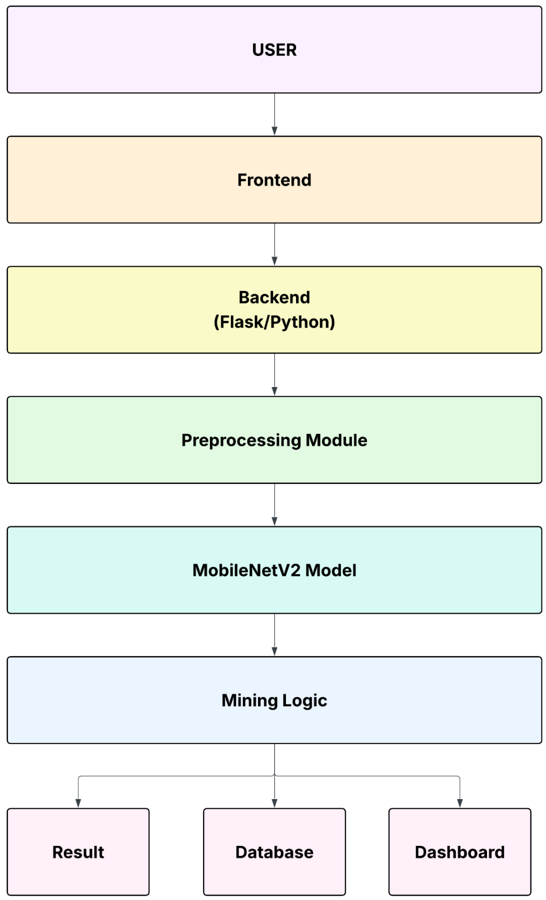
Tools Used:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Flask
- **Database:** SQLite
- **Model Serving:** TensorFlow / Keras

5.8 System Architecture

High-Level Architecture

User → Frontend → Flask Backend → Preprocessing Module → MobileNetV2 Model → Mining Logic → Result + Database + Dashboard



Module Description

- **Image Upload Module** – accepts image input
- **Preprocessing Module** – resizes and normalizes image
- **Prediction Module** – classifies image using CNN
- **Mining Detection Module** – converts class to mining / non-mining
- **Database Module** – stores result history
- **Dashboard Module** – displays analytics

6. Implementation

The implementation of the proposed system was carried out in Python using a combination of deep learning and web technologies.

6.1 Software and Tools Used

Programming Language: Python
Backend: HTML, CSS, Javascript
Deep Learning Framework: Flask
Image Processing: PIL, NumPy
Data Handling: Pandas
Evaluation: Scikit-learn, Matplotlib
Database: MySql

6.2 Dataset Handling

The **EuroSAT dataset** was used as the main image dataset. Images were organized class-wise and loaded into the training pipeline.

The classes were then interpreted under the mining approximation framework.

6.3 Model Development Steps

The implementation was completed in the following steps:

Step 1: Dataset Loading

Images were loaded from folders and labels were assigned automatically.

Step 2: Preprocessing

Images were:

- resized to 128×128,
- normalized,
- augmented.

Step 3: Model Building

MobileNetV2 was imported as the base model and additional classification layers were added.

Step 4: Training

The model was trained using training images and validated on validation data.

Step 5: Evaluation

Performance was checked using:

- Accuracy
- Loss
- Confusion Matrix
- Classification Report

Step 6: Flask Integration

The trained model was saved and loaded into a Flask application.

Step 7: Prediction and Storage

Uploaded images were classified and results were stored in SQLite.

Step 8: Dashboard and History

Prediction records were shown to users through a simple dashboard.

7. Results and Analysis

The developed system successfully classified satellite images and mapped selected land-use classes into mining / non-mining categories.

7.1 Output Behavior

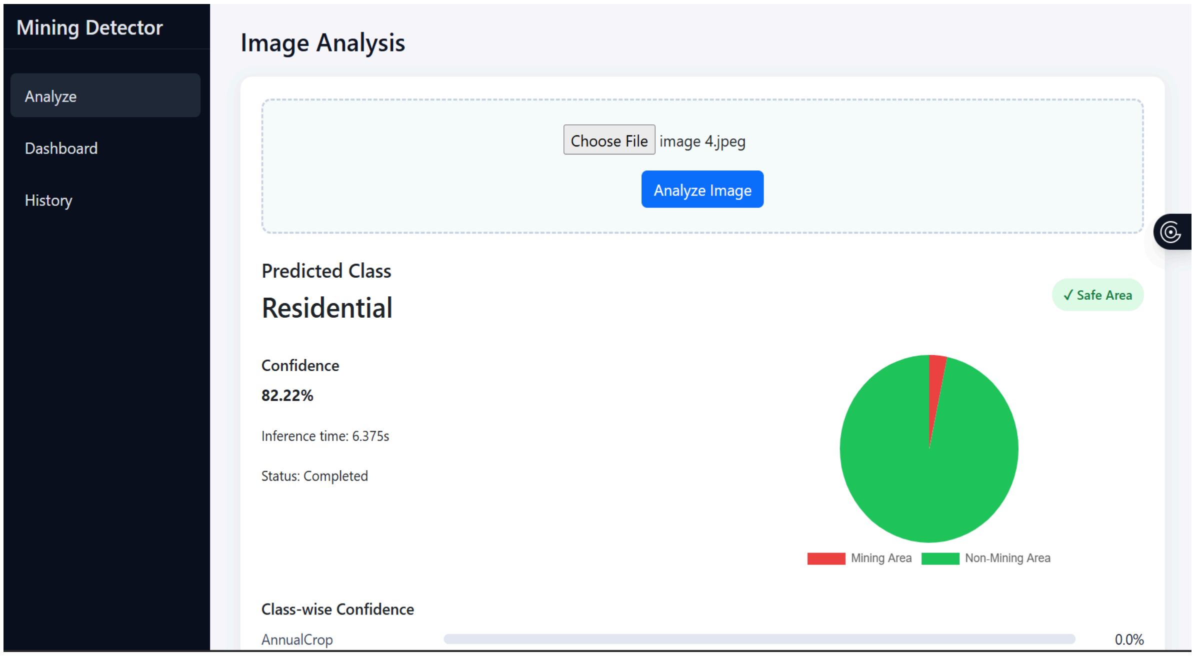
For each uploaded image, the system provides:

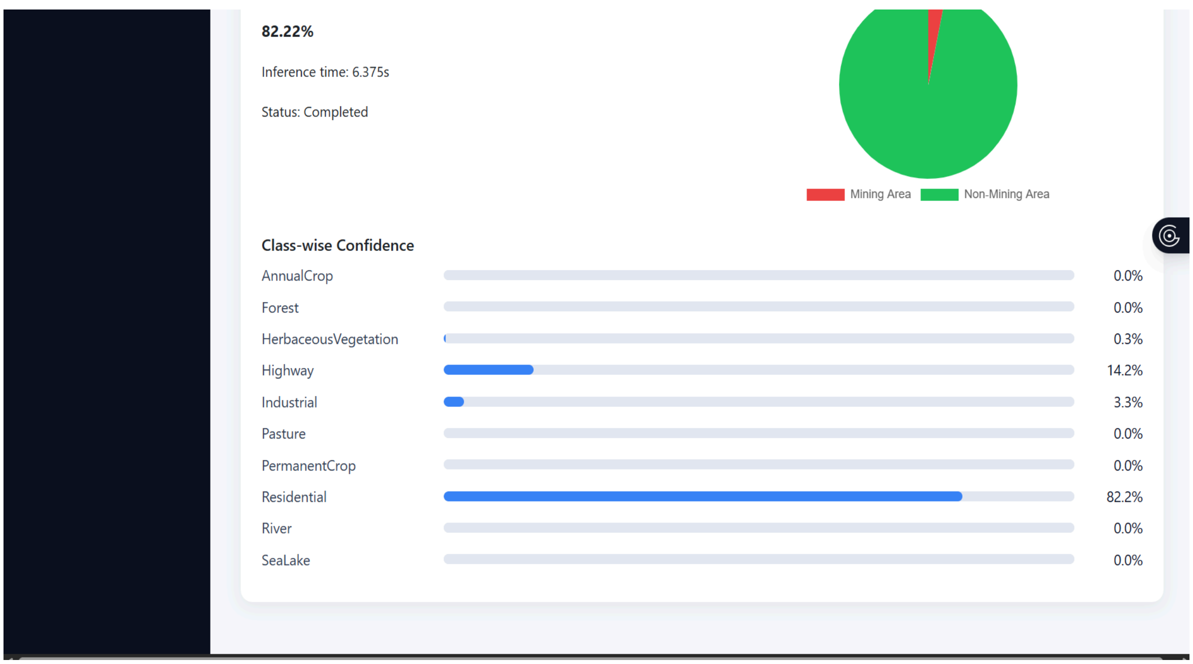
- predicted land class,
- confidence score,
- mining / non-mining result.

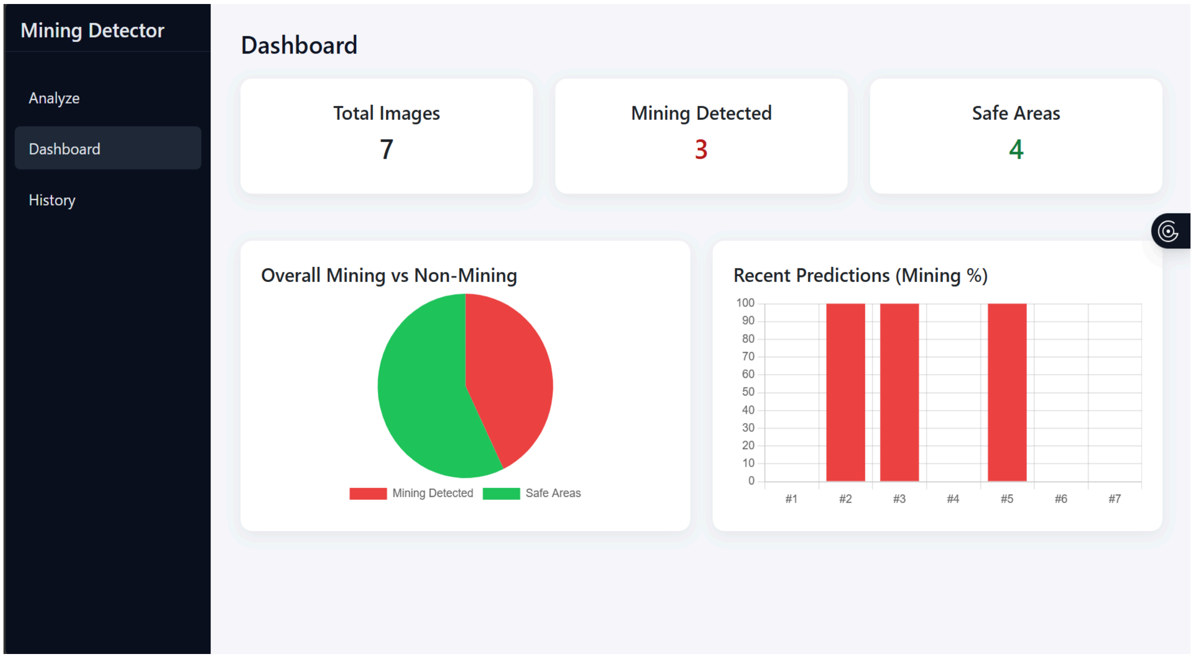
Example Outputs:

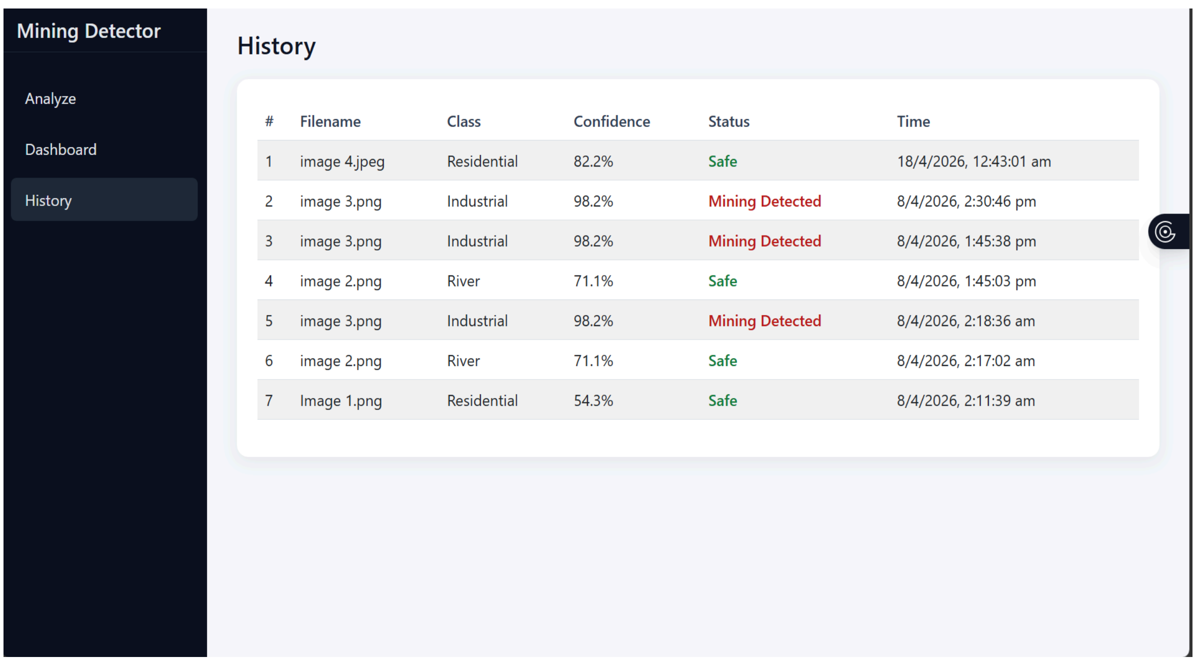
- Industrial → Mining Detected
- Forest → Non-Mining

- Residential → Non-Mining
- AnnualCrop → Mining Detected







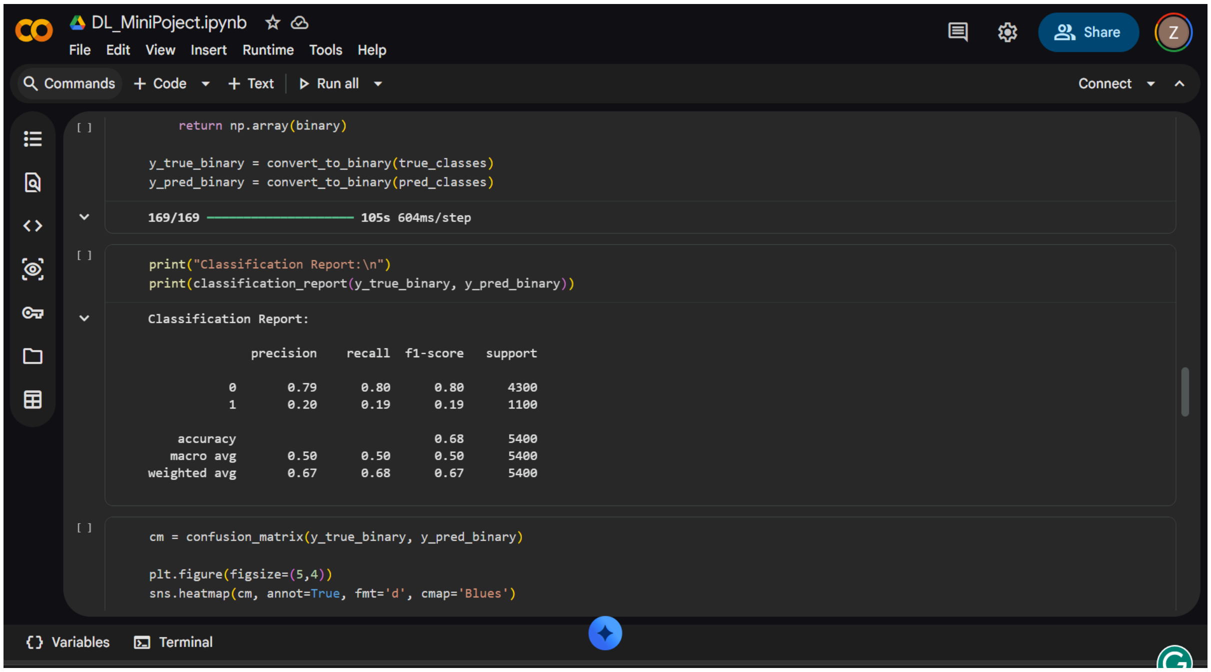


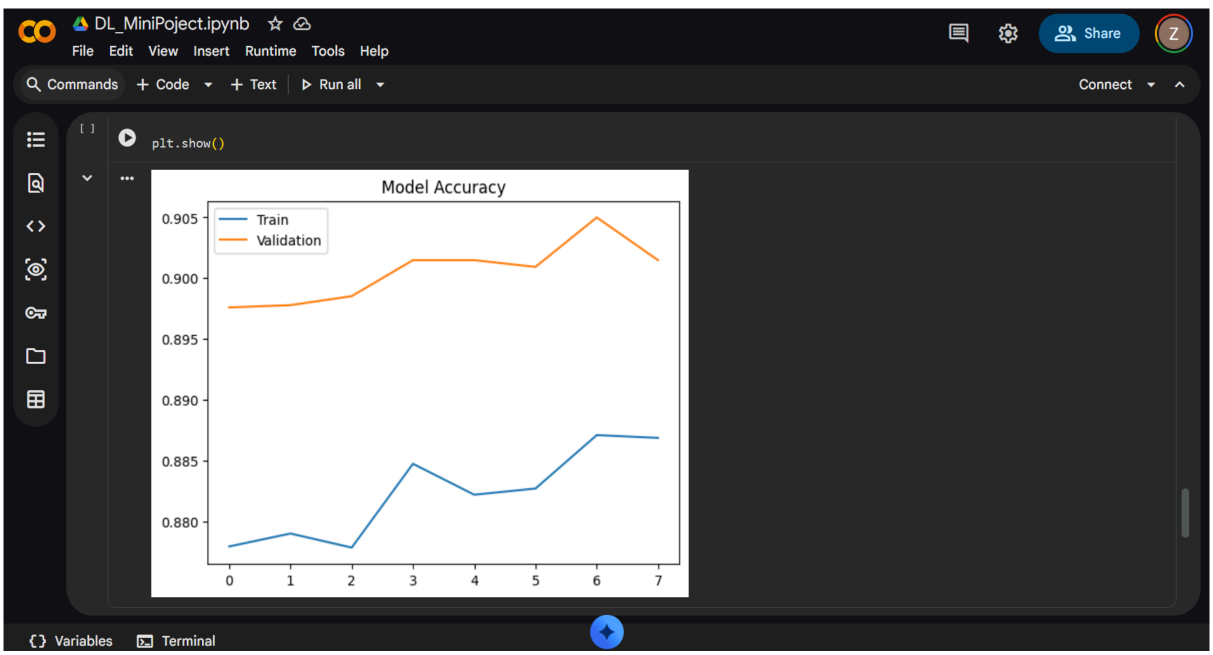
7.2 Evaluation Metrics

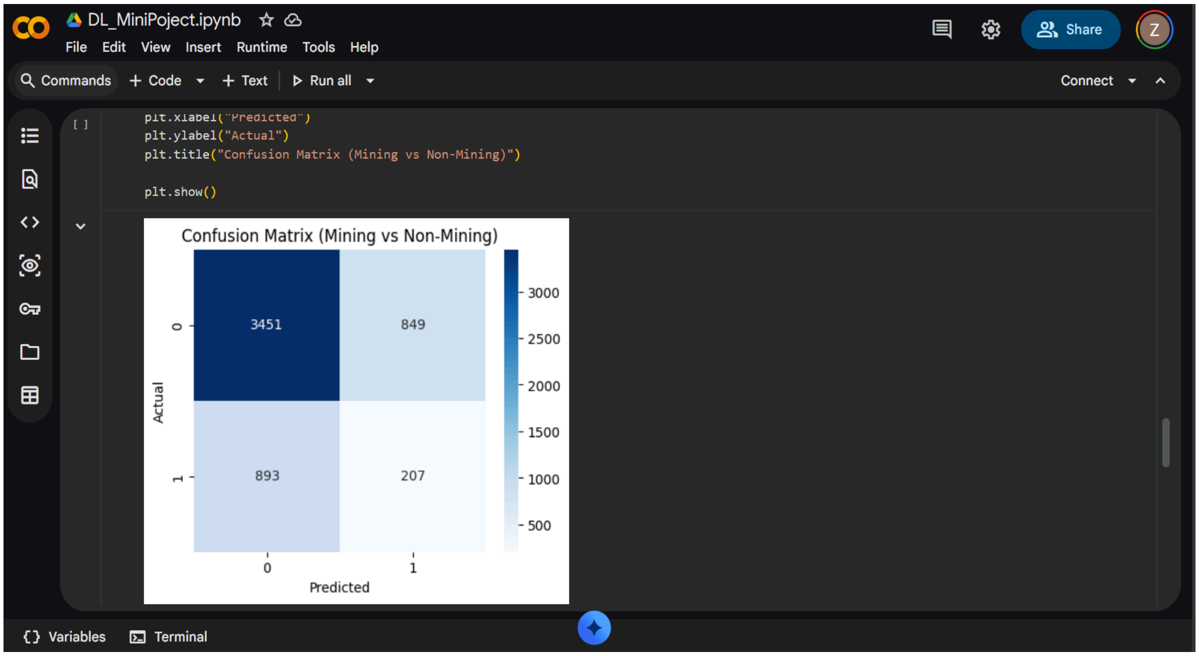
The performance of the model can be evaluated using:

- Accuracy
- Precision
- Recall

- F1-Score
- Confusion Matrix







7.3 Interpretation of Results

The model performs well for clearly distinguishable land-use classes such as:

- Industrial
- Forest
- River
- Residential

However, some classes with visually similar textures may show overlap, such as:

- AnnualCrop vs Pasture
- HerbaceousVegetation vs Forest

Even with this limitation, the model demonstrates that **satellite-image-based AI systems can effectively assist in identifying suspicious land-use patterns**.

➢ **Strength of the Result**

The major strength of this project is not just the model accuracy, but the fact that it provides a **working environmental monitoring prototype** that combines:

- image classification,
- mining approximation logic,

- frontend usability,
- and result storage.

That makes the project academically strong and practically demonstrable.

8. Discussion

The system developed in this project shows that detecting possible illegal surface mining using satellite imagery and deep learning is both feasible and practical, even with limited resources. Instead of depending on a specialised mining data set the project uses a smart approximation approach by analysing land use patterns and mapping them to mining related activities.

One important point to understand is that this system does not directly confirm whether mining is illegal or legal. It only identifies regions that look similar to mining areas based on satellite images. So, this system should be seen as a support tool for monitoring and screening rather than a final decision making system.

Despite this limitation the project successfully demonstrates how AI can simplify large-scale environmental monitoring tasks. It reduces the need for manual inspection and provides faster analysis of satellite data.

The integration of a user interface, database storage, and dashboard also makes the project more practical and usable compared to a basic model. It shows how machine learning models can be turned into real world applications.

Advantages

- Low-cost and scalable
- Uses publicly available data
- Lightweight model
- Easy to deploy
- Good academic value
- Can be extended further

Limitations

- No real-world illegal mining ground-truth labels
- Uses approximation logic
- Satellite image resolution may limit fine detection
- Cannot detect underground mining
- Cannot determine legality without external government mining records

Why the project is still valid

Even with these limitations, the project is still meaningful because:

- many real-world research papers also use **proxy detection, land-cover classification, segmentation, and remote sensing approximations**,
- it demonstrates technical feasibility,
- and it provides a solid foundation for future real-world systems.

9. Conclusion

This project successfully presents an AI based approach for identifying potential illegal surface mining areas using satellite imagery. By combining Deep learning techniques with remote sensing data, the system is able to classify land use patterns and highlight regions that may indicate mining activity.

The use of a tree trained MobileNetV2 model makes the system efficient and suitable for environments with limited power. Along with the model, the addition of a simple user interface prediction display and data storage overall usability of the system.

Although the project uses an approximation method due to the lack of dedicated mining dataset, it still demonstrates how an intelligent system can assist in environmental monitoring. It highlights the growing role of AI in solving Real world problems related to sustainability and resource management.

In summary, this project successfully shows that:

- satellite images can be effectively used for land analysis,
- CNNs can classify relevant surface patterns,
- and AI can support future illegal mining surveillance systems.

10. Future Scope

There are many ways this project can be improved and expanded in the future to make it more accurate than closer to real world applications.

Possible future enhancements:

1. Use a **real illegal mining dataset** instead of proxy land-use mapping.
2. Integrate **Sentinel-1 SAR imagery** for cloud-resistant detection.
3. Use **Sentinel-2 multispectral data** for better land analysis.

4. Replace image classification with **semantic segmentation** using **U-Net** for pixel-level mining area detection.
5. Use **YOLO-based object detection** for identifying mining equipment, roads, or excavation patches.
6. Add **change detection** to compare before/after satellite images.
7. Integrate **GIS maps** and geolocation tagging.
8. Connect with **government mining license databases** to detect truly illegal activity.
9. Add **real-time alert generation** for environmental agencies.
10. Deploy the system on cloud for large-scale regional monitoring.

This future scope makes your project look **serious, scalable, and research-worthy**.

11. References

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